

POLITECNICO
MILANO 1863

Digital Control Technology for Aeronautics
Academic Year 2025/2026

Design and validation of a digital control system for a flexible transmission

Person Code	Surname	Name
10767478	Cislaghi	Riccardo
10770085	Ahmed Mohamed Doubie	Karim

Contents

I Digital Controller Design for a Flexible Actuator **1**

1 Description of the Methodology Adopted and Design Criteria **1**
1.1 Methodology Adopted . . . 1
1.2 Justification of the θ_1/ω_m Cascade Architecture . . . 2
1.3 Design Criteria (Constraints Imposed by Performance Requirements) . . . 3

2 System Analysis and Z-Domain Requirements **4**
2.1 System Modeling and Controllability/Observability . . . 4
2.2 Structural Dynamics Analysis . . . 4

3 Justification for the Choice of Sampling Time **5**

4 Controller Design Process **5**
4.1 Initial Design Attempt and Rationale for Chosen Methodology . . . 5
4.2 Inner Speed Loop Controller ($C_{\text{speed}}(z)$) . . . 5
4.3 Outer Position Loop Controller ($C_{\text{pos}}(z)$) . . . 7

5 Evaluation of the Stability Margins of the Resulting System **7**

6 Performance Verification **8**
6.1 Performance Verification . . . 8
6.2 Robustness Analysis . . . 9

II Hybrid Simulation **9**

7 Signal acquisition **10**
7.1 Sensors . . . 10
7.2 Amplification and filtering . . . 10
7.2.1 Amplifier . . . 10
7.2.2 Anti aliasing filter . . . 11
7.3 Analog-to-Digital Conversion . . . 11
7.3.1 ADC . . . 11
7.3.2 DAC . . . 12
7.4 Final signal acquisition system . . . 12

8 Simulink simulation **12**

9 Performance evaluation **14**

10 Conclusions **15**

List of Tables

1 Summary of Controller Design Requirements . . . 3
2 Verification of Stability Margins . . . 7
3 Performance Verification Against Design Requirements . . . 8
4 Angular frequencies of the system . . . 11
5 Performance Verification of Hybrid Simulation Against Design Requirements . . . 15

List of Figures

1 Physical model of the flexible actuator with gearbox and elastic transmission. 1
2 Cascaded control structure with outer position loop and inner speed loop. 2
3 Bode diagram of the transfer functions ω_m/V_a and ω_1/V_a 2
4 Closed-loop disturbance rejection analysis of the full cascaded control architecture. The Bode diagram (left) illustrates the frequency-domain disturbance attenuation, while the step response (right) confirms zero steady-state deviation of the load position θ_1 and a finite motor offset due to elastic torque compensation. 3
5 Pole-Zero Map of the motor plant $G_{\text{motor}}(s)$, visually distinguishing the resonant poles from the anti-resonant zeros. 4
6 Open-Loop Inner Loop. 7
7 Open-Loop Outer Loop. 7
8 Step Response of the Final Outer Closed-Loop System $(\theta_l/\theta_{\text{ref}})$. 8
9 Task 1's Simulink implementation 8
10 Simulation results: load position tracking with disturbance rejection, actuator voltage, and load speed. 9
11 Bode magnitude and phase plot of the outer-loop sensitivity function $S_{pos}(z)$. 9
12 Sensor models in Simulink 10
13 Simulink model of the amplifier 11
14 Analog-to-Digital Converter Simulink model 12
15 Digital-to-Analog Converter 12
16 Signal acquisition system models in Simulink 12
17 Complete Simulink model 13
18 Simulink implementation of controllers 14
19 Step response of hybrid simulation 14
20 Position error of hybrid simulation 15

Part I

Digital Controller Design for a Flexible Actuator

Objective: The primary objective of this design phase is to develop and verify a discrete-time (Z-domain) control algorithm for a flexible aeronautical actuator. Part 1 considers an ideal digital control implementation. Sensors, filters, analog to digital conversion and vice versa are introduced in Part 2. The design process commences with the provided continuous-time model and must satisfy a set of rigorous performance requirements concerning transient response, steady-state accuracy, stability margins, and physical constraints.

1 Description of the Methodology Adopted and Design Criteria

1.1 Methodology Adopted

The system is identified as a fourth-order plant characterized by a lightly damped flexible mode, a classic example of a two-mass system with a flexible transmission. The standard and most robust method for controlling such systems is a cascade control architecture that ensures high-performance control by delegating specific tasks to progressively faster inner loops.

- **Theoretical Justification:** The full cascade architecture for a modern servomechanism consists of three loops: an outermost position loop, a middle speed loop, and an innermost current (torque) control loop. The innermost loop, typically implemented using Field-Oriented Control (FOC), is extremely fast (e.g., > 300 Hz) and is designed to behave as a high-bandwidth torque source. It ensures that the torque produced by the motor is directly proportional to the command it receives from the speed loop.
  Our design leverages this by treating the FOC loop as an ideal actuator. The presence of flexibility introduces a resonance that fundamentally limits the achievable bandwidth. By designing a fast middle loop to control the motor's speed, we stabilize the primary dynamics and create a simplified, well-behaved "speed actuator" for the slower outer loop to command. This bandwidth separation is crucial to ensure that the outer position controller operates well below the flexible resonance frequency, thereby avoiding excitation of the lightly damped mode.

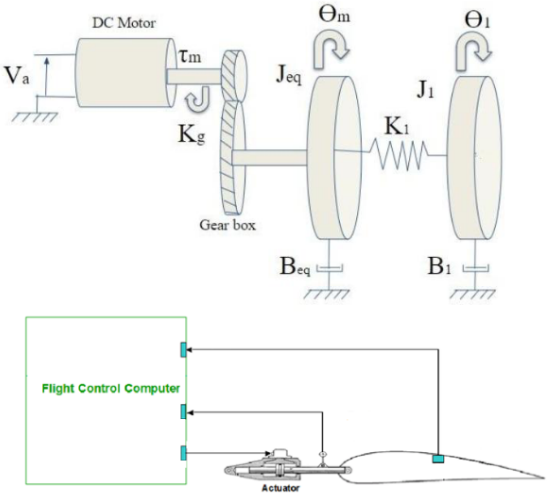


Figure 1: Physical model of the flexible actuator with gearbox and elastic transmission.

The chosen design methods for each loop are:

- **Inner Loop (Speed Control):** A fast inner loop is designed to control the motor speed (ω_m). This controller is synthesized using the Direct Synthesis (Ragazzini) method, an analytical technique that calculates the required controller based on a desired closed-loop response.
- **Outer Loop (Position Control):** A slower outer loop is designed to control the load position (θ_l). This controller is synthesized using classical loop shaping in the frequency domain, a robust method based on shaping the open-loop Bode plot to meet stability and bandwidth requirements.

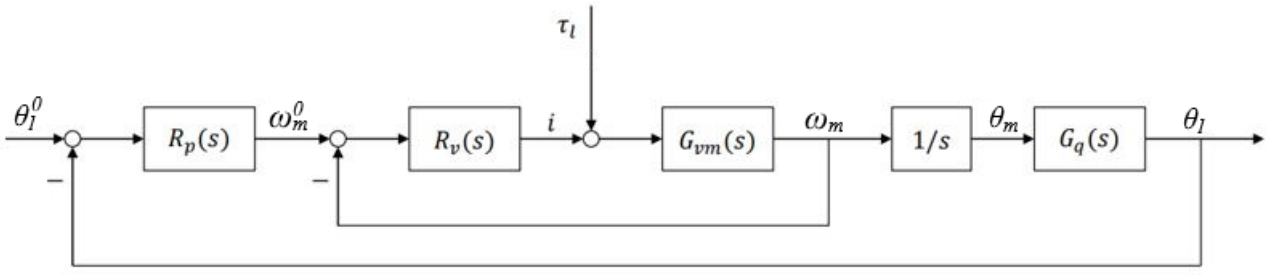


Figure 2: Cascaded control structure with outer position loop and inner speed loop.

1.2 Justification of the θ_1/ω_m Cascade Architecture

- **Inner Speed Loop Justification**

  The inner loop must regulate the actuator's fastest and most limiting dynamics in order to achieve the highest possible robust bandwidth.

  From a frequency-domain perspective, the Bode plot of the transfer functions ω_m/V_a and ω_1/V_a (Figure 3) reveals the characteristic anti-resonance/resonance pair of a flexible two-mass system. In particular, the response ω_m/V_a exhibits a pronounced anti-resonant notch around the flexible mode frequency, introducing a sharp magnitude drop and significant phase variation. This behavior imposes a practical upper bound on achievable bandwidth.

  By selecting ω_m as the feedback variable, the controller directly manages these limiting motor-side dynamics. In contrast, the transfer function ω_1/V_a exhibits additional phase lag due to the flexible transmission. Designing a high-bandwidth loop around ω_1 would therefore reduce robustness, since the motor-side flexible dynamics would still be excited but not directly regulated.

  Furthermore, the control input V_a acts directly on the motor torque and therefore directly influences ω_m. The path to ω_1 includes gearbox and shaft dynamics, introducing additional delay and phase lag. Closing the fast loop around ω_m, the variable most directly influenced by the actuator, maximizes achievable performance and stability.

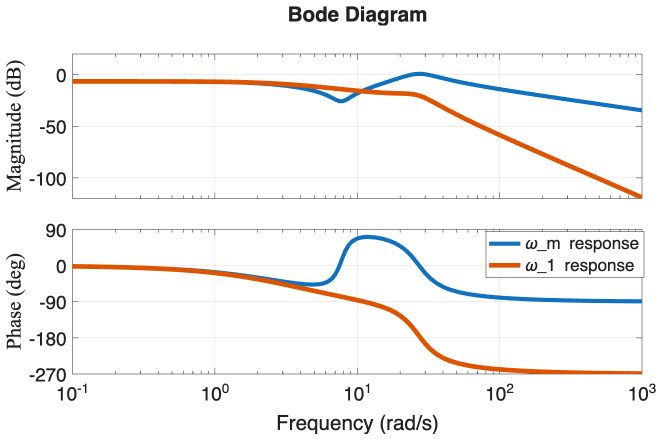


Figure 3: Bode diagram of the transfer functions ω_m/V_a and ω_1/V_a

- **Outer Position Loop Justification**

  The ultimate objective of the system is precise regulation of the load position θ_1.

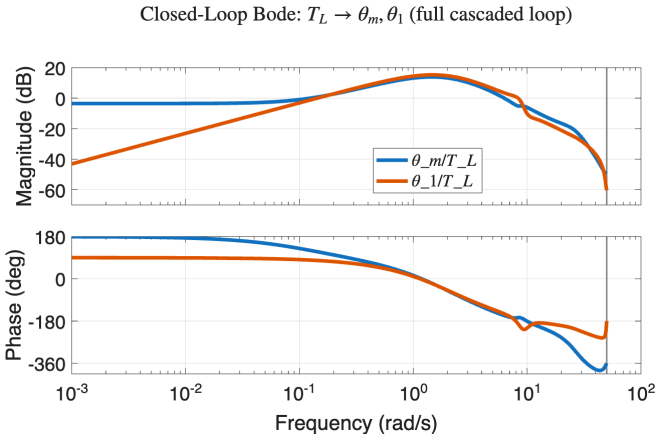
  Under a constant load torque disturbance T_L, the flexible shaft undergoes elastic deformation, resulting in $\theta_m \neq \theta_1$. The Bode plot of the closed loop disturbance transfer functions θ_m/T_L and θ_1/T_L (Figure 4a and Figure 4b) highlights the different disturbance propagation characteristics between motor and load sides.

  If the outer loop were closed on θ_m, the torsional deformation of the shaft would translate into a steady-state error at the load. By instead using θ_1 as feedback, the controller compensates for elastic wind-up and ensures accurate regulation of the true controlled output.

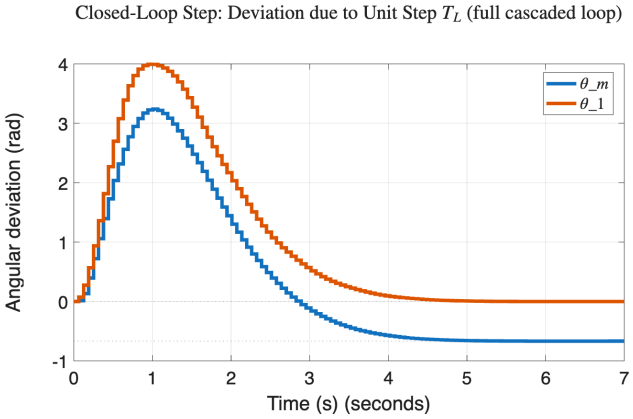
  Moreover, since the plant integrates speed to position,

$$\theta_1(s) = \frac{1}{s}\,\omega_1(s),$$

  the outer-loop plant is inherently Type-1. This guarantees zero steady-state error for step position commands and rejection of constant load torque disturbances without requiring an explicit integrator in the position controller.



(a) Closed-loop Bode plot: $T_L \to \theta_m, \theta_1$.



(b) Closed-loop step response to a unit step torque disturbance.

Figure 4: Closed-loop disturbance rejection analysis of the full cascaded control architecture. The Bode diagram (left) illustrates the frequency-domain disturbance attenuation, while the step response (right) confirms zero steady-state deviation of the load position θ_1 and a finite motor offset due to elastic torque compensation.

1.3 Design Criteria (Constraints Imposed by Performance Requirements)

The controller design is driven by the following project requirements, summarized in Table 1.

Table 1: Summary of Controller Design Requirements

Category	**Specification**
Transient Response	Overshoot must be less than 10%.
Steady-State Accuracy	Error for constant references must be $< 0.3°$.
Robustness	Phase Margin $> 45°$ for all control loops.
Disturbance Rejection	Must reject constant load torque disturbances.
Performance Goal	Achieve the fastest possible response with minimal oscillation.
Implementation	Sampling rate must be as low as possible.
Physical Limits	Actuator Voltage: $\pm$12V Output Axis Speed: $\pm 45°/s$

2 System Analysis and Z-Domain Requirements

2.1 System Modeling and Controllability/Observability

The system's physical parameters are defined in MATLAB to construct the continuous-time state-space model. A preliminary analysis confirms the system is fully controllable and observable (i.e., $y = x$). This is a critical theoretical prerequisite, guaranteeing that a stabilizing controller can be designed to place the closed-loop poles at any desired location.

$$\begin{bmatrix} \dot{\theta}_m(t) \\ \ddot{\theta}_m(t) \\ \dot{\theta}_1(t) \\ \ddot{\theta}_1(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1}{J_{eq}} & -\frac{B_{eq}}{J_{eq}} & \frac{\eta_m \eta_g K_g^2 K_t K_m}{R_a J_{eq}} & \frac{K_1}{J_{eq}} \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{J_1} & 0 & -\frac{K_1}{J_1} & -\frac{B_1}{J_1} \end{bmatrix}}_{A_{\text{full}}} \begin{bmatrix} \theta_m(t) \\ \dot{\theta}_m(t) \\ \theta_1(t) \\ \dot{\theta}_1(t) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{\eta_m \eta_g K_g K_t}{R_a J_{eq}} & 0 \\ 0 & 0 \\ 0 & \frac{1}{J_1} \end{bmatrix}}_{B_{\text{full}}} \begin{bmatrix} V_a(t) \\ T_L(t) \end{bmatrix} \quad (1)$$

2.2 Structural Dynamics Analysis

The transfer function from the control input V_a to the motor speed ω_m, $G_{\text{motor}}(s)$, is derived, as it represents the plant for the crucial inner control loop. The analysis of its poles and zeros reveals the system's core dynamic challenges.

- **Analysis:**
  - **Resonance (Poles):** The system has a resonant frequency at $\omega_p \approx 25.8$ rad/s. This is caused by the pair of lightly damped complex conjugate poles arising from the flexible coupling between the motor inertia and the load inertia. This resonance is the primary source of unwanted oscillations.
  - **Anti-Resonance (Zeros):** The $G_{\text{motor}}(s)$ transfer function has an anti-resonant frequency at $\omega_z \approx 7.68$ rad/s. This pair of complex conjugate zeros near to the imaginary axis represents the fundamental performance limitation of the system as it is the frequency at which the motor and load oscillate out of phase, causing the control input (motor torque) to have minimal effect on the motor's own speed. Any attempt to design a control loop with a bandwidth near or above this frequency will be ineffective, require excessive control effort, and likely lead to instability.

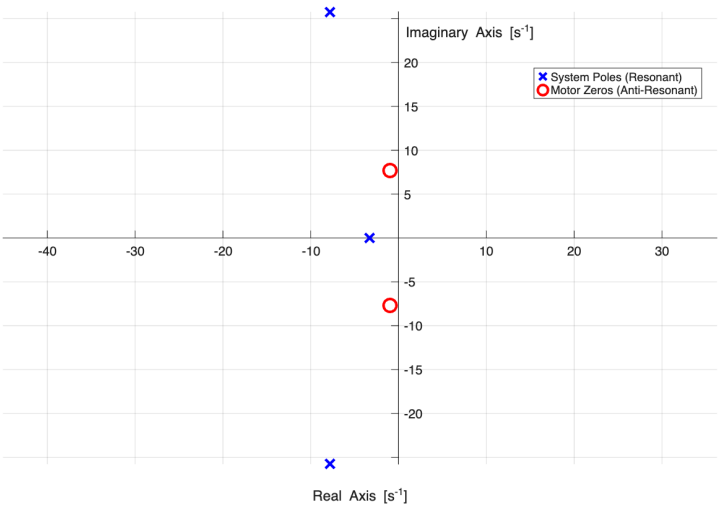


Figure 5: Pole-Zero Map of the motor plant $G_{\text{motor}}(s)$, visually distinguishing the resonant poles from the anti-resonant zeros.

3 Justification for the Choice of Sampling Time

The "fastest possible response" is fundamentally limited by the anti-resonance frequency ω_z.

- **Bandwidth Selection:** To ensure stability and avoid exciting the anti-resonance, the target bandwidth for the inner speed loop must be chosen to be safely below this frequency. A value of $\omega_{\text{speed_loop}} = 5$ rad/s is selected.
- **Sampling Time Calculation:** Based on the common heuristic rule $f_s \approx (10 - 20) \times f_{\text{loop}}$ to ensure a good discrete-time approximation of continuous behavior, a factor of 20 is chosen.

$$T_s = \frac{2\pi}{20 \cdot \omega_{\text{speed_loop}}} = 0.0628 \text{ s} \quad \text{(Sampling Frequency } f_s \approx 15.92 \text{ Hz)}.$$

4 Controller Design Process

4.1 Initial Design Attempt and Rationale for Chosen Methodology

The initial approach to designing the inner speed loop controller was to use a classical method with a standard Proportional-Integral (PI) structure. This was attempted using MATLAB's interactive `sisotool` environment to graphically shape the root locus and Bode plots of the discretized plant $G_{\text{motor}}(z)$.

- **Attempted Methodology:** A PI controller, $C(z) = K_p + K_i \frac{T_s}{z-1}$, was introduced. The goal was to place the controller's zero to cancel one of the plant's poles and then adjust the gain K_p to meet bandwidth and phase margin requirements.
- **Observed Result and Theoretical Failure:** While it was possible to achieve the target phase margin and bandwidth for the dominant system behavior, the simulation of the closed-loop step response revealed unsatisfactory performance. Specifically, the response exhibited persistent, high-frequency oscillations around the steady-state value.
- **Theoretical Reasoning for Failure:** The limitation of the PI controller arises from its restricted shaping capability when applied to a flexible two-mass plant. The resonance/anti-resonance pair introduces a pronounced magnitude peak and a rapid phase drop around the flexible mode. As a result, a PI controller cannot simultaneously (i) achieve the desired crossover frequency with adequate phase margin and (ii) sufficiently attenuate the resonant peak (i.e., provide enough damping of the lightly-damped flexible poles). Consequently, the closed-loop response remains strongly influenced by the flexible mode, which manifests as persistent oscillations in the time-domain step response. This motivates the adoption of a more structured synthesis approach (e.g., Ragazzini direct synthesis) to enforce a well-damped target closed-loop dynamics.

4.2 Inner Speed Loop Controller ($C_{\text{speed}}(z)$)

The continuous plant $G_{\text{motor}}(s)$ is discretized using the MATLAB command `Gz_motor = c2d(G_motor, Ts, 'zoh')`. The 'zoh' method is an exact discretization based on the physical behavior of a digital-to-analog converter (DAC).

- **Theoretical Reasoning:** The method assumes the continuous-time plant is preceded by a Zero-Order Hold (ZOH), which holds the control signal constant between sampling instants. The `c2d` function calculates the discrete-time state-space matrices (A_d, B_d) from the continuous ones (A_c, B_c) using the exact analytical solution:

$$A_d = e^{A_c T_s}$$
$$B_d = \left(\int_0^{T_s} e^{A_c \tau} d\tau \right) B_c$$

This ensures that the resulting discrete model $G(z)_motor$ exactly matches the continuous system's response at the sampling instants kT_s.

- **Formula:** The resulting discrete-time transfer function of the motor-side speed dynamics is

$$G(z)_{\text{motor}}(z) = \frac{0.4491z^2 - 0.7181z + 0.3951}{z^3 - 0.7551z^2 + 0.3278z - 0.3056}.$$

The controller $C_{\text{speed}}(z)$ is calculated analytically using the Ragazzini method. This method inverts the model of the closed-loop system to directly solve for the controller that achieves a specified target response $F^*(z)$.

- **Theoretical Reasoning:** The standard closed-loop transfer function is $F(z) = \frac{C(z)G(z)}{1+C(z)G(z)}$. Algebraically solving for the controller $C(z)$ yields the Ragazzini formula:

$$\begin{cases} C_{\text{speed}}(z) = \dfrac{1}{G(z)} \dfrac{F^*(z)}{1 - F^*(z)} \\ \\ C_{\text{speed}}(z) = \dfrac{0.04013z^6 - 0.05639z^5 + 0.006714z^4 + 0.01629z^3 - 0.01371z^2 + 0.01368z - 0.005306}{0.4491z^6 - 2.048z^5 + 3.971z^4 - 4.176z^3 + 2.495z^2 - 0.7974z + 0.1056} \end{cases}$$

- **Design Choices and Verification of Theoretical Constraints:** To ensure the resulting controller $C(z)$ is both causal and stable, the choice of the target closed-loop transfer function $F^*(z)$ must adhere to three critical rules which arise from the structure of this equation. These rules were verified as follows:
  - **Target Response Selection:** The target response was first defined in the continuous domain as a standard, critically damped ($\zeta = 1$) second-order system to ensure a smooth response with no overshoot, directly addressing the project requirements.
    - Bandwidth: $\omega_n = \omega_{\text{speed_loop}} = 5$ rad/s.
    - Final Form: $F_s(s) = \frac{25}{s^2+10s+25}$.
    - Discretization: The Zero-Order Hold ('zoh') method was chosen to discretize $F_s(s)$ into $F^*(z)$, as it exactly models the physical behavior of the system's DAC.
  - **Rule 1: Causality Constraint.** For the controller to be physically realizable, its output cannot depend on future inputs. This imposes a condition on the relative degree (poles - zeros) of the system:

$$\text{relative degree}(F^*(z)) \geq \text{relative degree}(G_{\text{motor}}(z))$$

    - **Verification:** The plant $G_{\text{motor}}(z)$ is a third-order system with a relative degree of one. The chosen second-order target response $F^*(z)$ also has a relative degree of 1. The causality constraint is therefore satisfied.
  - **Rule 2: Stability Constraint (Handling Non-Minimum Phase Zeros).** The controller term $1/G(z)$ attempts to cancel the zeros of the plant. To prevent the cancellation of an unstable (non-minimum phase) zero, which would lead to internal instability,all non-minimum phase (NMP) zeros of the plant $G(z)$ must also be zeros of the target response $F^*(z)$.
    - **Verification:** An analysis of the discretized plant, $G_{\text{motor}}(z)$, confirmed that all its zeros are located inside the unit circle. The plant is therefore **minimum-phase**, and this constraint is trivially satisfied without needing to modify $F^*(z)$.

– **Rule 3: Stability Constraint (Handling Unstable Poles).** Similarly, the term in the denominator of the controller, $(1 - F^*(z))$, must not introduce an unstable pole by attempting to cancel an unstable pole from the plant, that's why all unstable poles of the plant $G(z)$ must be zeros of the term $1 - F^*(z)$.

  * **Verification:** An analysis confirmed that all poles of the plant $G_{\text{motor}}(z)$ are located inside the unit circle. The plant is therefore **stable**, and this constraint is also trivially satisfied without modification to $F^*(z)$.

– **Practical Consideration: Robustness.** The Ragazzini method relies on **perfect cancellation** of the plant's dynamics by the controller.

4.3 Outer Position Loop Controller ($C_{\text{pos}}(z)$)

A simple proportional controller $C_{\text{pos}}(z) = K_p$ is designed for the outer loop.

- **Theoretical Reasoning:** The plant for the outer loop, $G_{\text{pos_plant}}$, is Type 1 because it includes the discrete integrator $\frac{T_s}{z-1}$ that converts load speed to load position. For a Type 1 system, a proportional controller is sufficient to achieve a zero steady-state error for a step input. The gain K_p is then used to set the desired bandwidth and stability margins.
- **Gain Selection:** An initial estimate for the gain was calculated as $K_p = 1/|G(j\omega_c)|$ using a target crossover frequency of $\omega_c = 0.9$ rad/s (such that the bandwidth separation ratio greater than 5). To meet the transient performance requirements, this gain was fine-tuned via simulation to a final value of $K_p = 0.77$, which successfully reduced the system's overshoot. As a result of this gain adjustment, the final crossover frequency for the outer loop was reduced to 0.76 rad/s.

5 Evaluation of the Stability Margins of the Resulting System

The stability of both control loops is verified by analyzing their open-loop Bode plots.Both loops comfortably exceed the $> 45°$ phase margin requirement, indicating a robust design.

Table 2: Verification of Stability Margins

Control Loop	Phase Margin
Inner Speed Loop	74.2°
Outer Position Loop	67.2°

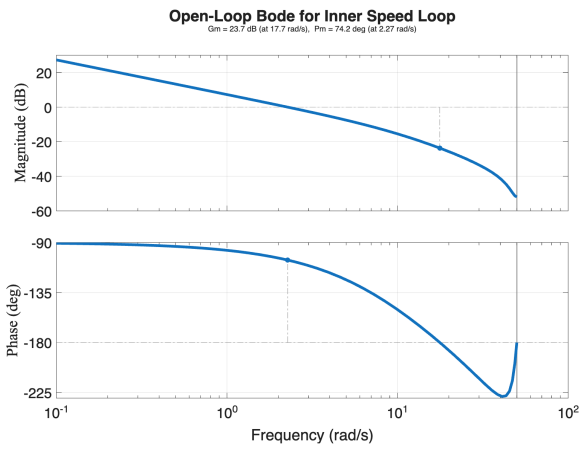


Figure 6: Open-Loop Inner Loop.

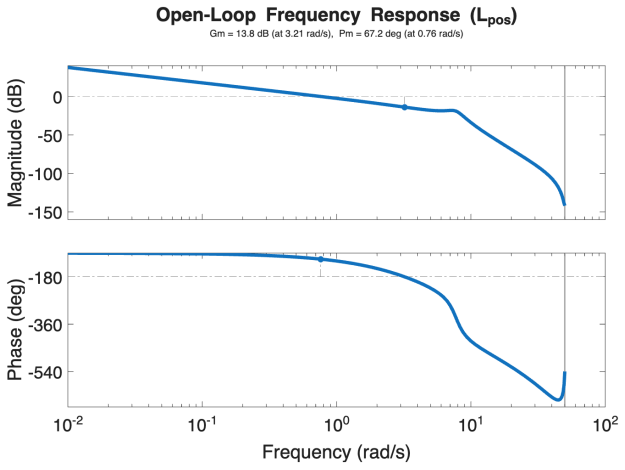


Figure 7: Open-Loop Outer Loop.

6 Performance Verification

6.1 Performance Verification

The performance of the final (outer) closed-loop system is demonstrated in the step response shown in Figure 8. A detailed analysis confirms that all design requirements are met, as summarized in Table 3.

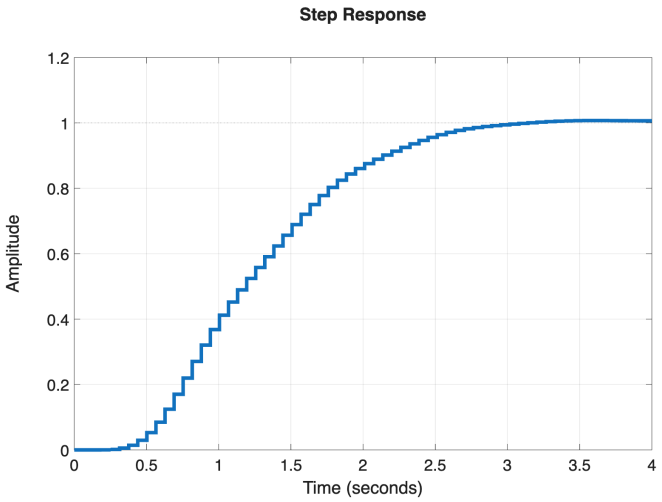


Figure 8: Step Response of the Final Outer Closed-Loop System ($\theta_l/\theta_{\text{ref}}$).

Table 3: Performance Verification Against Design Requirements

Requirement	**Result from Simulation**	**Status**
Overshoot $< 10\%$	0.56%	✓ Pass
Settling Time	2.76 seconds	✓ Pass
Steady-State Error $< 0.3°$	Zero steady-state error	✓ Pass
Disturbance Rejection	Disturbance is fully rejected	✓ Pass
Actuator Voltage $< \pm 12$V	Peak voltage is ≈ 2.2V	✓ Pass
Output Speed $< \pm 45°/$s	Peak speed is $\approx 43.5°/$s (0.7585 rad/s)	✓ Pass

Analysis of Simulation Results:

- **Disturbance Rejection:** A step disturbance of $T_L = 0.05$ Nm (applied at 6s) is effectively rejected. This is a direct result of the integral action present in the system, which ensures the open-loop gain at DC is infinite.
- **Constraint Compliance:** Simulation plots confirm that both the actuator voltage ($V_a(t)$) and the load speed ($\omega_l(t)$) remain well within their specified operational limits.

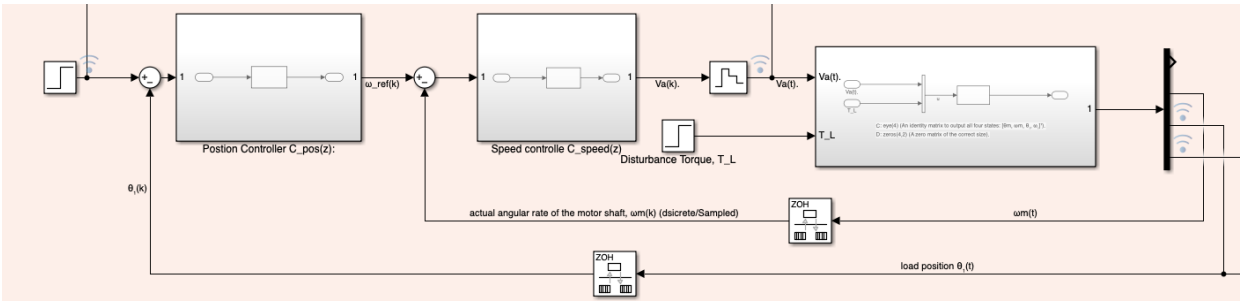


Figure 9: Task 1's Simulink implementation

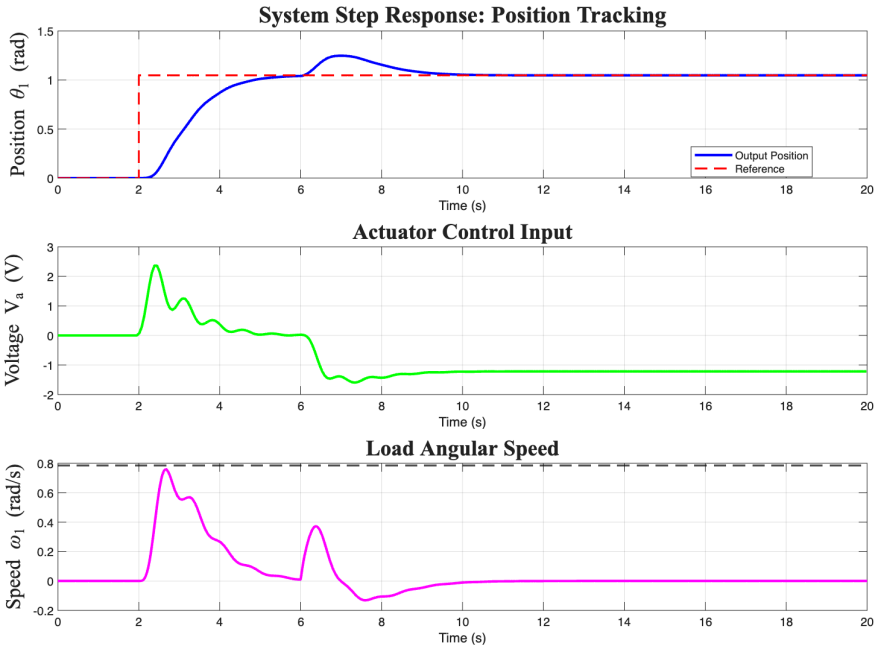


Figure 10: Simulation results: load position tracking with disturbance rejection, actuator voltage, and load speed.

6.2 Robustness Analysis

The sensitivity function

$$S_{pos}(z) = \frac{1}{1 + L_{pos}(z)},$$

describes the closed-loop system's response to disturbances and modeling uncertainties, and provides a key indicator of closed-loop robustness.

- The magnitude remains significantly below 0 dB at low frequencies, confirming excellent rejection of constant and slow-varying disturbances, consistent with the time-domain simulations. On the other hand at high frequencies, the sensitivity magnitude approaches 0 dB, as expected, since disturbance attenuation is sacrificed beyond the closed-loop bandwidth.
- Around the crossover region, the sensitivity function exhibits a limited peak (approximately 3–4 dB), corresponding to a maximum sensitivity value $M_s = \|S(z)\|_\infty \approx 1.5$. This moderate peak confirms acceptable robustness margins while preserving strong low-frequency disturbance rejection.

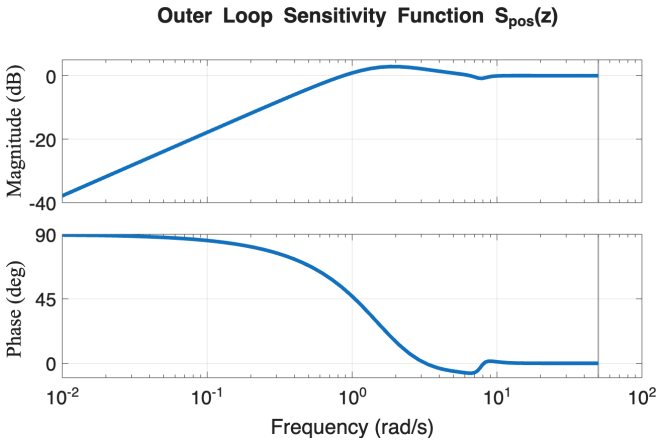


Figure 11: Bode magnitude and phase plot of the outer-loop sensitivity function $S_{pos}(z)$.

Part II

Hybrid Simulation

After designing the controllers, a hybrid simulation taking into consideration the effects of signal measurement has been developed in Simulink.

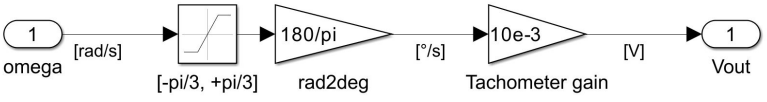
7 Signal acquisition

7.1 Sensors

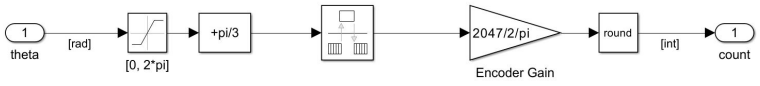
Both motor and load axes possess tachometers to measure angular velocity and optical encoders to measure angular displacement. The former can measure angular velocities within $\pm$ 60°/s, with a sensitivity of 10 $mV/(°/s)$ and 0V output for zero-velocity; the output voltage thus lies within $\pm$ 0.6V.

The latter behave differently, as they have a resolution of 2048 counts per revolution, so the output is not a voltage but directly an integer number, thus power amplification and Analog-to-Digital conversion will not be necessary for this kind of sensor. Since the output shaft rotation angle is limited within $\pm$ 60° and the output of the encoder is an integer number ranging from 0 to 2047, a bias is introduced in the sensor's model such that a measurement of 0 is obtained with a rotation of $-60°$ and 2047 with a rotation of $+60°$. In reality this bias can be obtained by simply aligning the zero of the encoder with the proper axis rotation.

Figure 12 shows how the sensors have been implemented in Simulink; saturation blocks have been used to limit the range of the measured variable, also note that the simulation works with angles expressed in radians.



(a) Tachometer



(b) Encoder

Figure 12: Sensor models in Simulink

7.2 Amplification and filtering

7.2.1 Amplifier

As stated above the tachometer output lies within $\pm 0.6V$, while the Analog-to-Digital Converter (ADC), which will be described in the following section, accepts an input voltage in the range $[0V, 3.3V]$, so an amplifier is needed to ensure proper conversion of the measurement. The amplifier follows the equation $y = ax + b$, so in order to map the interval $[-0.6V \quad +0.6V]$ to $[0V \quad 3.3V]$ a gain of 2.75 and a bias of 1.65 are needed. The Simulink implementation is thus straightforward as shown in figure 13.

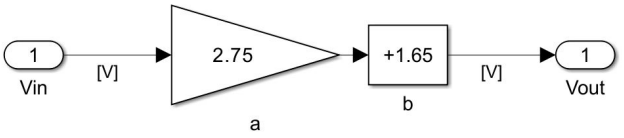


Figure 13: Simulink model of the amplifier

7.2.2 Anti aliasing filter

Since the controller works in discrete time, an analog anti-aliasing filter is needed before signal sampling. The filter consists of a low-pass filter, defined by its cut-off frequency and order. Since no information is given in regards to sensor noise, a simple first order filter was chosen, although if more noise attenuation is needed a higher order one can be used, at the cost of less phase margin. The transfer function of such filter is:

$$G_{filter}(s) = \frac{\omega_c}{s + \omega_c}$$

The cut-off frequency ω_c needs to be higher than the closed-loop bandwidth to avoid excessive phase margin loss due to the filter, but it also has to be lower than the Nyquist frequency to avoid aliasing. After many considerations it was chosen to be slightly more than half the Nyquist frequency for the inner loop, and for the outer loop 20 times the closed-loop bandwidth. Out of the two, the one for the inner loop was the hardest to select, as lower values led to higher overshoot in the speed loop, which in the final simulation led the angular velocity to exceed 45 °/s; in the end it was found that the lowest acceptable value is 27 rad/s. The table below lists important angular frequencies of the system.

Angular frequency ω	**Value [rad/s]**
Sampling	100
Nyquist	50
Inner loop bandwidth	5
Inner loop filter	27
Outer loop bandwidth	0.77
Outer loop filter	15.4

Table 4: Angular frequencies of the system

7.3 Analog-to-Digital Conversion

Since the real system is in continuous time and the controllers work in discrete time, it is necessary to implement an Analog-to-Digital Converter (ADC) and vice versa (DAC).

7.3.1 ADC

The key parameter in sizing the ADC is the number of bits, which could be chosen between 8 and 12. The final choice landed on 8, as it guarantees good performance while keeping the ADC cheaper.

With 8 bits the full range of measurement from the tachometer is divided in 255 possible values, thus the maximum error on angular velocity stemming from conversion is:

$$e_\omega = \frac{1}{2} \cdot \frac{\omega_{max} - \omega_{min}}{2^{n_{bit}} - 1} = 0.2353°/s$$

Propagating this error for one sampling period yields a position error of 0.0148°, which is much lower than the required 0.3° tracking error. Furthermore, the discretization error caused by the encoder is:

$$e_\theta = \frac{1}{2} \cdot \frac{360°}{2047} = 0.088°$$

which is almost six times greater, thus we can conclude that 8 bits are more than enough to guarantee the desired accuracy.

Additionally the ADC has a conversion delay of $1\mu s$, although its effects are negligible in the closed loop behavior as it is much smaller than the sampling time, which is equal to 62.8 ms. Figure 14 shows the implementation of the ADC in Simulink.

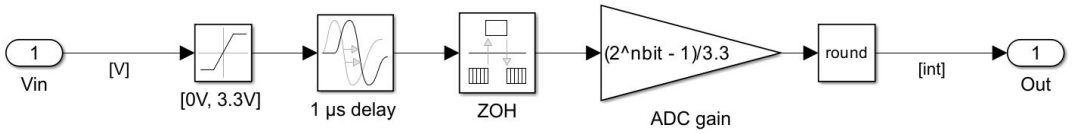


Figure 14: Analog-to-Digital Converter Simulink model

7.3.2 DAC

The DAC takes as input an integer value between 0 and 1023, so it works with 10 bits, and outputs a voltage between -12V and +12V. Its implementation in Simulink is straightforward, as it consists only of a gain and a bias, with saturation added to limit the output, without any conversion delay. The Simulink model is depicted in figure 15.

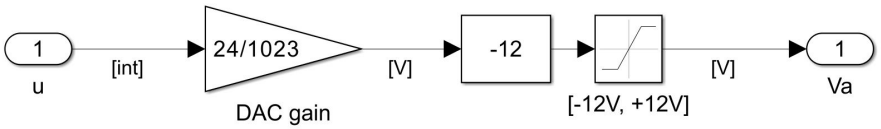
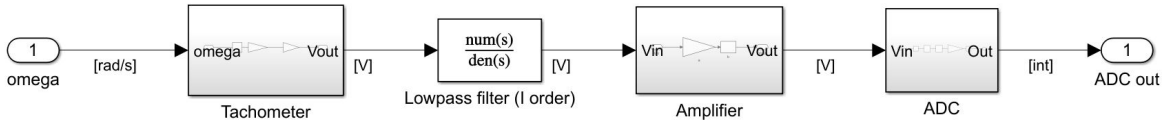


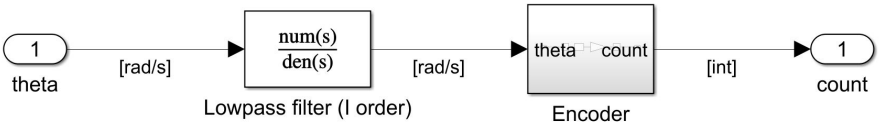
Figure 15: Digital-to-Analog Converter

7.4 Final signal acquisition system

Putting together what has been said, we come to the following implementation of signal acquisition systems for both loops.



(a) Inner loop



(b) Outer loop

Figure 16: Signal acquisition system models in Simulink

8 Simulink simulation

The full hybrid simulation Simulink model is depicted in figure 17.

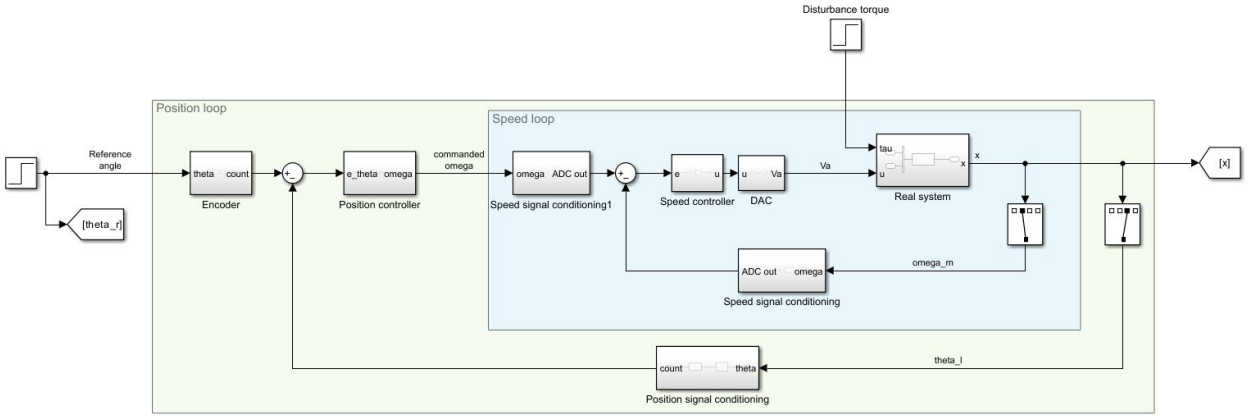


Figure 17: Complete Simulink model

One thing to note is that the output of the signal acquisition systems and the input of the DAC are all dimensionless integer numbers, so it is necessary to also transform the reference value of each loop. Additionally it is necessary to compensate the overall gain and bias of the signal acquisition + DAC system inside of the controllers, since both have been designed with the real variables as input-output: the speed controller expects the actual angular velocity as input and motor voltage as output, while the position controller expects the actual shaft position as input and angular velocity as output.

In both cases the input bias is compensated by applying the signal conditioning to both the reference and measured signals before subtracting them (the anti-aliasing filter is omitted for the reference signals). The only source of bias to be compensated for is the DAC, where the range [0, 1023] gets converted to [-12V, +12V], so a bias of +512 is applied to output of the speed controller.

Meanwhile the overall gain has to be compensated inside of the controller themselves. For the speed controller the signal acquisition gain is:

$$G_{signal} = \frac{2^{n_{bit}} - 1}{\omega_{max} - \omega_{min}} = 2.125\ s/^\circ = 121.8\ s/rad$$

and the DAC gain is:

$$G_{DAC} = \frac{V_{max} - V_{min}}{1023} = 0.235\ V$$

so the overall gain to compensate for in the controller is:

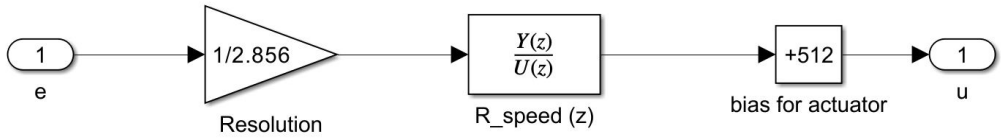
$$G_{total} = G_{signal}\ G_{DAC} = 0.05\ V/^\circ = 2.856\ V/rad$$

The output of the position controller, on the other hand, doesn't go to a DAC, so the only gain that needs to be compensated is the signal acquisition one, which is:

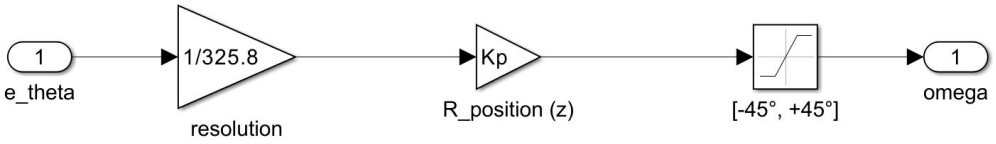
$$G_{signal} = G_{total} = \frac{2047}{360^\circ} = 5.686/^\circ = 325.8/rad$$

Since it is requested for the angular velocity not to exceed $45^\circ/s$, the output of the controller is saturated to such values.

In the real control system all of this would be implemented directly in the controller, but in our Simulink model it has been implemented as shown in figure 18.



(a) Speed controller



(b) Position controller

Figure 18: Simulink implementation of controllers

9 Performance evaluation

The results from the simulation are reported in figure 19; comparing it to the step response obtained in the design part, figure 10, it can be seen that the two are almost identical, so the effect of sampling and conversion is minimal.

From the first graph it can also be seen why it is necessary to use the output shaft position in the outer loop instead of the motor shaft position: because of the external torque applied at the output and the flexible transmission, the motor shaft needs to be at a different angle than the output at steady state.

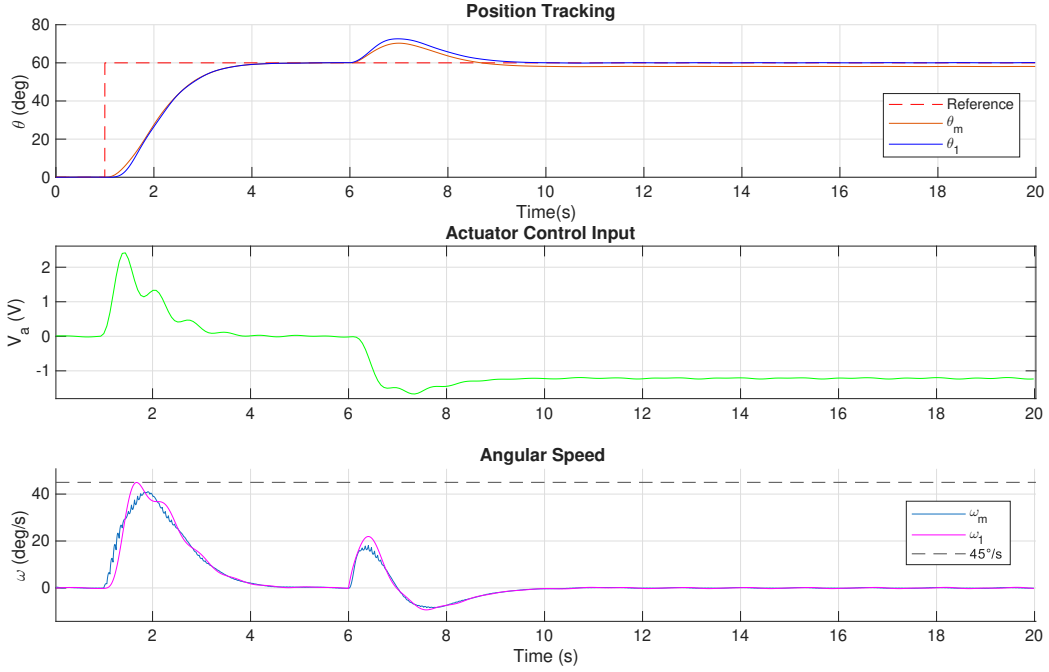


Figure 19: Step response of hybrid simulation

Furthermore the error between the desired and actual output shaft position is plotted in figure 20, and it can be seen that after the transient is over it is always below the threshold of 0.3°.

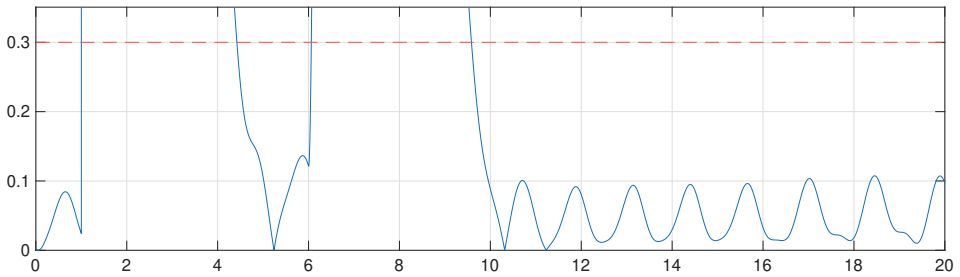


Figure 20: Position error of hybrid simulation

Since the hybrid simulation's behavior is very similar to what predicted in the design phase, it is clear that the requirements are all verified.

Table 5: Performance Verification of Hybrid Simulation Against Design Requirements

Requirement	**Result from Simulation**	**Status**
Overshoot $< 10\%$	0.56%	✓ Pass
Settling Time	2.76 seconds	✓ Pass
Steady-State Error $< 0.3°$	Zero steady-state error	✓ Pass
Disturbance Rejection	Disturbance is fully rejected	✓ Pass
Actuator Voltage $< \pm 12$V	Peak voltage is ≈ 2.4V	✓ Pass
Output Speed $< \pm 45°/s$	Peak speed is $\approx 44.95°/s$ (0.7645 rad/s)	✓ Pass

10 Conclusions

A discrete-time cascading control architecture with inner loop designed through direct synthesis, also called Ragazzini's method, and outer loop controller designed with loop shaping techniques proved to be an effective solution to control the position of a flexible transmission, satisfying all of the design requirements. The control system was verified through the development of a hybrid simulation taking into account sensors, filters and conversion from analog to digital and vice versa.